

Mona Tarek Mohamed

AI Engineer

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SUMMARY

AI Engineer with hands-on experience building end-to-end LLM and AI systems, including RAG pipelines, agentic workflows, and machine learning solutions. Experienced in technical research, experimentation, evaluation, and developing production-ready AI services with Python, APIs, and modern AI frameworks.

EDUCATION

Bachelor's Degree in Electronics and Communication Engineering

2021 – 2025 | Alexandria, Egypt

Faculty of Engineering, Alexandria University
CGPA: 3.13

EXPERIENCE

LLM Fine-Tuning Research Assistant | Alexandria University

07/2026 – Present

Research Project under the Supervision of Dr. Ahmed Eltrass

- Investigating advanced LLM fine-tuning and post-training techniques, including supervised fine-tuning, parameter-efficient methods, reinforcement learning with GRPO, and preference optimization.
- Implementing and experimenting with Llama-based models across reasoning and question-answering tasks, with hands-on work in dataset preparation, training configuration, PEFT, quantization, and GPU-based experimentation.
- Exploring and evaluating reward-driven training and LLM evaluation, including correctness and structured-format rewards and LLM-as-a-Judge evaluation.

AI Engineer

02/2026 – 06/2026 | Alexandria, On-site

Rockai Dev - Full time

- Built production AI systems, including autonomous agents, multi-step LLM workflows, and RAG pipelines, focusing on structured outputs, modularity, reliability, and evaluation.
- Developed end-to-end AI pipelines spanning prompt engineering, vector retrieval, agent orchestration, structured LLM outputs, and API integration.
- Evaluated and debugged AI workflows by identifying retrieval, generation, and output-validation issues, improving system reliability and consistency.

Generative AI Engineer Intern

01/2025 – 02/2025 | Online

Sprints - Internship

- Applied advanced prompt engineering in production, optimizing prompts to improve accuracy, reduce hallucinations, and enhance human-AI interaction.
- Planned and implemented an AI agent with a RAG pipeline, including document processing, embeddings, vector indexing, retrieval, and RAG optimization.

Artificial Intelligence Intern

07/2024 – 08/2024 | Alexandria, On-site

Telecom Egypt - Internship

- Developed machine learning solutions for classification, regression, and clustering using Scikit-learn, including Random Forest, SVM, XGBoost, Logistic Regression, Linear Regression, and Gradient Boosting.
- Built data pipelines and feature engineering workflows, evaluating model performance using cross-validation, performance metrics, regularization, and hyperparameter tuning to improve generalization and robustness.

SELECTED PROJECTS

AI-Powered Python Code Tutor & Compiler

Python, FastAPI, LLMs, Pydantic, AST, Docker

- Built an AI-powered Python code execution and review platform with an end-to-end pipeline for code validation, execution, prompt construction, LLM analysis, and structured feedback.
- Developed AST-based security and validation checks to diagnose and prevent execution failures, including syntax errors, restricted imports, and unsafe operations, alongside FastAPI APIs and a chatbot interface.

Reyada | Agentic Startup Journey & Market Research Agent

Python, LangChain, LangSmith, LLMs, Tavily, MongoDB, FastAPI

- Built a ReAct agent using LangChain, integrated with Tavily web search and LangSmith tracing, enabling observable multi-step market research workflows for startup founders.

- Designed a multi-phase agent architecture with structured LLM outputs, an LLMFactory abstraction supporting multiple providers, strict schema validation, MongoDB persistence, and FastAPI integration.

Voice Command Mini-RAG System (Developed as part of Graduation Project)

Linux, Python, Vosk, ONNX, ChromaDB, Docker, Raspberry Pi

- Developed a low-latency offline speech-to-text pipeline using a local Hugging Face model on Linux, enabling on-device intelligence and responsive interaction for edge AI environments with sub-3-second response times.
- Built a mini RAG-based voice command system by integrating ChromaDB as a vector database for semantic command retrieval, and deployed it as a Dockerized service on a Raspberry Pi, enabling privacy-preserving, hands-free control for ADAS systems.

SKILLS

Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, Scikit-learn, Neural Networks, Model Optimization, Hyperparameter Tuning, Transfer Learning, Model Evaluation

Generative AI & LLMs: RAG, Agents, LangChain, LangSmith, Prompt Engineering, Transformers, AI Pipeline Development

Data Storage & Retrieval: Pandas, NumPy, MongoDB, Vector Databases (ChromaDB, Qdrant), Embeddings, Vector Retrieval

Software Engineering & Deployment: Python, OOP, Git, Linux | FastAPI, API development, Docker, Streamlit